

Reproducibility supplement

Do MOS Fusion Weights Select Useful Embeddings? A System-Disjoint Attribution Audit

Daniel Casanova, Alef Iury Ferreira, Edresson Casanova, Lucas Gris, Pedro Lustosa Rege Botelho, Fernanda Silva, Anderson da Silva Soares

Artifact version: v1.0-icassp2027-submission. Repository: <https://github.com/danielcasanova12/mos-fusion-weight-audit>. The release tag resolves the immutable Git commit.

This document accompanies the machine-readable CSVs and reports all 40 attribution cells, all 16 discordant cells with uncertainty and refit frequency, all 120 Rashomon coordinate intervals, system-disjoint selection by corpus and budget, and the null controls. Values are copied from the canonical committed CSVs; the numbered tables and supplementary CSVs identify the evidence used by the paper.

Data and pretrained checkpoints are not redistributed. See `configs/` and `checkpoints/README.md` for sources, versions, licenses, and preprocessing instructions. The public artifact begins from already extracted representations.

This compact release is a frozen results package: the numerical CSVs and this PDF are the publication artifacts. Private extraction/fitting drivers, audio, ratings, and licensed feature caches are outside the public snapshot.

Table S1 — All attribution cells (1/2)

Gate is the full-coalition allocation. Singleton, LOO, and Shapley are held-out MSE utilities; positive means lower MSE. Train +freq is the fraction of training-system refits with positive Shapley. Test CI is a system bootstrap with fitted models fixed.

Corpus	Expert	Gate	Singleton	LOO	Shapley	CI low	CI high	Shapley +freq (50)
BRSpeechMOS	BEATs	6.36e-18	0.3614	-7.28e-10	0.0488	-0.0111	0.1088	0.5000
BRSpeechMOS	CED	0	0.5884	3.28e-08	0.1243	0.0644	0.1839	0.9800
BRSpeechMOS	ContentVec	0.1532	0.2955	0.0238	0.0885	0.0030	0.1801	0.8800
BRSpeechMOS	ERB	0	0.4721	-1.01e-09	0.0514	0.0143	0.0908	0.6400
BRSpeechMOS	RMVPE-c	1.39e-17	-0.0065	9.01e-09	0.0042	-0.0352	0.0416	0.9800
BRSpeechMOS	RMVPE-q	2.81e-17	-0.0776	6.97e-08	-0.0033	-0.0191	0.0237	0.3800
BRSpeechMOS	Speaker	1.80e-16	0.5518	-9.53e-08	0.1202	0.0826	0.1753	1.0000
BRSpeechMOS	WavLM	0.3021	0.4876	0.0238	0.1295	0.0485	0.2125	0.9200
BRSpeechMOS	Whisper	0.0298	-0.3430	-0.0048	-0.0486	-0.1261	0.0476	0.3800
BRSpeechMOS	eGeMAPS	0.5149	0.6512	0.0152	0.1693	0.0585	0.2870	1.0000
BVCC	BEATs	0.2003	0.5651	0.0106	0.1137	0.0896	0.1402	1.0000
BVCC	CED	0.2222	0.5682	0.0139	0.1158	0.0909	0.1415	1.0000
BVCC	ContentVec	0.0614	0.4625	0.0023	0.0794	0.0576	0.1027	1.0000
BVCC	ERB	3.70e-17	0.1782	-9.36e-08	0.0232	0.0118	0.0374	1.0000
BVCC	RMVPE-c	2.35e-17	0.0503	-1.04e-07	0.0068	0.0010	0.0131	1.0000
BVCC	RMVPE-q	0	0.0211	-1.01e-07	0.0034	-0.0019	0.0091	0.5200
BVCC	Speaker	0.0994	0.5004	0.0014	0.0858	0.0629	0.1104	1.0000
BVCC	WavLM	0.1672	0.5347	0.0016	0.0939	0.0701	0.1190	1.0000
BVCC	Whisper	0.2495	0.4067	-0.0024	0.0656	0.0430	0.0894	1.0000
BVCC	eGeMAPS	0	0.3386	-9.87e-09	0.0500	0.0334	0.0684	1.0000

Table S1 — All attribution cells (2/2)

Gate is the full-coalition allocation. Singleton, LOO, and Shapley are held-out MSE utilities; positive means lower MSE. Train +freq is the fraction of training-system refits with positive Shapley. Test CI is a system bootstrap with fitted models fixed.

Corpus	Expert	Gate	Singleton	LOO	Shapley	CI low	CI high	Shapley +freq (50)
SingMOS	BEATs	0.1525	0.0983	0.0014	0.0375	0.0159	0.0598	0.9800
SingMOS	CED	0.2314	0.1725	0.0109	0.0605	0.0347	0.0859	1.0000
SingMOS	ContentVec	0.0761	-0.0919	-0.0025	-0.0005	-0.0214	0.0195	0
SingMOS	ERB	1.01e-17	0.0418	-1.54e-08	0.0181	0.0066	0.0290	1.0000
SingMOS	RMVPE-c	7.00e-18	0.0092	2.87e-08	0.0111	0.0047	0.0177	1.0000
SingMOS	RMVPE-q	4.21e-18	-0.0681	5.89e-08	0.0027	-0.0109	0.0143	0.2400
SingMOS	Speaker	0.0078	0.1052	0.0002	0.0324	0.0232	0.0411	1.0000
SingMOS	WavLM	0.2156	0.0782	0.0037	0.0321	0.0142	0.0495	1.0000
SingMOS	Whisper	0.1036	-0.2350	0.0030	-0.0113	-0.0520	0.0255	0
SingMOS	eGeMAPS	0.2129	0.1738	0.0117	0.0579	0.0418	0.0736	1.0000
TMHINT-QI	BEATs	0.2710	0.5989	0.0058	0.1035	0.0543	0.1709	1.0000
TMHINT-QI	CED	0.2837	0.5920	0.0002	0.0964	0.0466	0.1656	1.0000
TMHINT-QI	ContentVec	0.0716	0.5263	0.0003	0.0805	0.0327	0.1477	1.0000
TMHINT-QI	ERB	0	0.1861	3.51e-09	0.0222	0.0086	0.0388	1.0000
TMHINT-QI	RMVPE-c	3.08e-17	0.0293	4.94e-09	0.0023	0.0008	0.0040	0.8800
TMHINT-QI	RMVPE-q	7.52e-19	0.0292	2.74e-09	0.0032	-0.0027	0.0093	0.1800
TMHINT-QI	Speaker	0.0118	0.4944	-1.19e-06	0.0724	0.0334	0.1266	1.0000
TMHINT-QI	WavLM	0.1697	0.6062	0.0065	0.1064	0.0580	0.1736	1.0000
TMHINT-QI	Whisper	0.1922	0.5903	0.0091	0.1040	0.0605	0.1646	1.0000
TMHINT-QI	eGeMAPS	0	0.4858	2.20e-08	0.0695	0.0255	0.1323	1.0000

Table S2 — Zero-gate / positive-Shapley cells (1/1)

The full set of 16 discordant cells. Gate zero uses the predeclared threshold $<1e-6$. Gate-0 freq (100) refits preprocessing, experts, OOF predictions, and the gate under training-system resampling; Shapley +freq (50) recomputes the complete coalition game in 50 such refits. The separate frozen-expert bootstrap is archived but is not shown in this table.

Corpus	Expert	Gate	LOO	Shapley	CI low	CI high	Gate-0 freq (100)	Shapley +freq (50)
BRSpeechMOS	BEATs	6.36e-18	-7.28e-10	0.0488	-0.0111	0.1088	0.4700	0.5000
BRSpeechMOS	CED	0	3.28e-08	0.1243	0.0644	0.1839	0.5400	0.9800
BRSpeechMOS	ERB	0	-1.01e-09	0.0514	0.0143	0.0908	0.8300	0.6400
BRSpeechMOS	RMVPE-c	1.39e-17	9.01e-09	0.0042	-0.0352	0.0416	0.6900	0.9800
BRSpeechMOS	Speaker	1.80e-16	-9.53e-08	0.1202	0.0826	0.1753	0.4600	1.0000
BVCC	ERB	3.70e-17	-9.36e-08	0.0232	0.0118	0.0374	1.0000	1.0000
BVCC	RMVPE-c	2.35e-17	-1.04e-07	0.0068	0.0010	0.0131	1.0000	1.0000
BVCC	RMVPE-q	0	-1.01e-07	0.0034	-0.0019	0.0091	1.0000	0.5200
BVCC	eGeMAPS	0	-9.87e-09	0.0500	0.0334	0.0684	0.9400	1.0000
SingMOS	ERB	1.01e-17	-1.54e-08	0.0181	0.0066	0.0290	0.7300	1.0000
SingMOS	RMVPE-c	7.00e-18	2.87e-08	0.0111	0.0047	0.0177	0.9300	1.0000
SingMOS	RMVPE-q	4.21e-18	5.89e-08	0.0027	-0.0109	0.0143	0.5400	0.2400
TMHINT-QI	ERB	0	3.51e-09	0.0222	0.0086	0.0388	1.0000	1.0000
TMHINT-QI	RMVPE-c	3.08e-17	4.94e-09	0.0023	0.0008	0.0040	1.0000	0.8800
TMHINT-QI	RMVPE-q	7.52e-19	2.74e-09	0.0032	-0.0027	0.0093	0.9700	0.1800
TMHINT-QI	eGeMAPS	0	2.20e-08	0.0695	0.0255	0.1323	0.2900	1.0000

Table S3 — Complete Rashomon intervals (1/4)

Coordinatewise minimum and maximum gate weights among mixtures whose OOF MSE is within the stated relative tolerance of the optimum. All solver-success flags are checked by `verify_release.py`; KKT diagnostics are in `results/rashomon_kkt_audit.csv`.

Corpus	Expert	Tol.	Opt.	Min	Max	Width	Min OK	Max OK
BRSpeechMOS	BEATs	0.0010	0	0	0.0019	0.0019	yes	yes
BRSpeechMOS	CED	0.0010	1.67e-16	1.67e-16	0.0157	0.0157	yes	yes
BRSpeechMOS	ContentVec	0.0010	0.1532	0.1225	0.1839	0.0614	yes	yes
BRSpeechMOS	ERB	0.0010	0	0	0.0009	0.0009	yes	yes
BRSpeechMOS	RMVPE-c	0.0010	1.08e-16	1.08e-16	0.0069	0.0069	yes	yes
BRSpeechMOS	RMVPE-q	0.0010	2.44e-16	2.44e-16	0.0017	0.0017	yes	yes
BRSpeechMOS	Speaker	0.0010	0	0	0.0207	0.0207	yes	yes
BRSpeechMOS	WavLM	0.0010	0.3021	0.2685	0.3357	0.0672	yes	yes
BRSpeechMOS	Whisper	0.0010	0.0298	0.0049	0.0547	0.0498	yes	yes
BRSpeechMOS	eGeMAPS	0.0010	0.5149	0.4790	0.5503	0.0713	yes	yes
BVCC	BEATs	0.0010	0.2003	0.1609	0.2397	0.0788	yes	yes
BVCC	CED	0.0010	0.2222	0.1783	0.2661	0.0878	yes	yes
BVCC	ContentVec	0.0010	0.0614	0.0228	0.1000	0.0772	yes	yes
BVCC	ERB	0.0010	9.51e-18	9.51e-18	0.0028	0.0028	yes	yes
BVCC	RMVPE-c	0.0010	0	0	0.0033	0.0033	yes	yes
BVCC	RMVPE-q	0.0010	1.82e-18	1.82e-18	0.0030	0.0030	yes	yes
BVCC	Speaker	0.0010	0.0994	0.0591	0.1397	0.0806	yes	yes
BVCC	WavLM	0.0010	0.1672	0.1244	0.2100	0.0856	yes	yes
BVCC	Whisper	0.0010	0.2495	0.2137	0.2853	0.0716	yes	yes
BVCC	eGeMAPS	0.0010	0	0	0.0075	0.0075	yes	yes
SingMOS	BEATs	0.0010	0.1525	0.1139	0.1912	0.0773	yes	yes
SingMOS	CED	0.0010	0.2314	0.1831	0.2796	0.0965	yes	yes
SingMOS	ContentVec	0.0010	0.0761	0.0440	0.1083	0.0643	yes	yes
SingMOS	ERB	0.0010	4.68e-18	4.68e-18	0.0176	0.0176	yes	yes
SingMOS	RMVPE-c	0.0010	1.13e-17	1.13e-17	0.0083	0.0083	yes	yes
SingMOS	RMVPE-q	0.0010	4.92e-18	4.92e-18	0.0162	0.0162	yes	yes
SingMOS	Speaker	0.0010	0.0078	0	0.0534	0.0534	yes	yes
SingMOS	WavLM	0.0010	0.2156	0.1716	0.2597	0.0881	yes	yes
SingMOS	Whisper	0.0010	0.1036	0.0733	0.1339	0.0606	yes	yes
SingMOS	eGeMAPS	0.0010	0.2129	0.1606	0.2632	0.1026	yes	yes

Table S3 — Complete Rashomon intervals (2/4)

Coordinatewise minimum and maximum gate weights among mixtures whose OOF MSE is within the stated relative tolerance of the optimum. All solver-success flags are checked by `verify_release.py`; KKT diagnostics are in `results/rashomon_kkt_audit.csv`.

Corpus	Expert	Tol.	Opt.	Min	Max	Width	Min OK	Max OK
TMHINT-QI	BEATs	0.0010	0.2710	0.1899	0.3521	0.1622	yes	yes
TMHINT-QI	CED	0.0010	0.2837	0.2002	0.3672	0.1670	yes	yes
TMHINT-QI	ContentVec	0.0010	0.0716	2.20e-15	0.1442	0.1442	yes	yes
TMHINT-QI	ERB	0.0010	6.75e-19	6.75e-19	0.0107	0.0107	yes	yes
TMHINT-QI	RMVPE-c	0.0010	8.99e-18	8.99e-18	0.0118	0.0118	yes	yes
TMHINT-QI	RMVPE-q	0.0010	9.74e-18	9.74e-18	0.0123	0.0123	yes	yes
TMHINT-QI	Speaker	0.0010	0.0118	2.67e-16	0.0769	0.0769	yes	yes
TMHINT-QI	WavLM	0.0010	0.1697	0.0844	0.2550	0.1707	yes	yes
TMHINT-QI	Whisper	0.0010	0.1922	0.1244	0.2599	0.1355	yes	yes
TMHINT-QI	eGeMAPS	0.0010	0	0	0.0449	0.0449	yes	yes
BRSpeechMOS	BEATs	0.0050	0	0	0.0092	0.0092	yes	yes
BRSpeechMOS	CED	0.0050	1.67e-16	1.67e-16	0.0552	0.0552	yes	yes
BRSpeechMOS	ContentVec	0.0050	0.1532	0.0846	0.2218	0.1372	yes	yes
BRSpeechMOS	ERB	0.0050	0	0	0.0043	0.0043	yes	yes
BRSpeechMOS	RMVPE-c	0.0050	1.08e-16	1.08e-16	0.0317	0.0317	yes	yes
BRSpeechMOS	RMVPE-q	0.0050	2.44e-16	2.44e-16	0.0081	0.0081	yes	yes
BRSpeechMOS	Speaker	0.0050	0	0	0.0669	0.0669	yes	yes
BRSpeechMOS	WavLM	0.0050	0.3021	0.2269	0.3772	0.1503	yes	yes
BRSpeechMOS	Whisper	0.0050	0.0298	9.58e-16	0.0854	0.0854	yes	yes
BRSpeechMOS	eGeMAPS	0.0050	0.5149	0.4235	0.5940	0.1705	yes	yes
BVCC	BEATs	0.0050	0.2003	0.1122	0.2884	0.1762	yes	yes
BVCC	CED	0.0050	0.2222	0.1240	0.3204	0.1964	yes	yes
BVCC	ContentVec	0.0050	0.0614	3.37e-15	0.1477	0.1477	yes	yes
BVCC	ERB	0.0050	9.51e-18	9.51e-18	0.0132	0.0132	yes	yes
BVCC	RMVPE-c	0.0050	0	0	0.0157	0.0157	yes	yes
BVCC	RMVPE-q	0.0050	1.82e-18	1.82e-18	0.0142	0.0142	yes	yes
BVCC	Speaker	0.0050	0.0994	0.0093	0.1896	0.1803	yes	yes
BVCC	WavLM	0.0050	0.1672	0.0715	0.2629	0.1914	yes	yes
BVCC	Whisper	0.0050	0.2495	0.1694	0.3296	0.1601	yes	yes
BVCC	eGeMAPS	0.0050	0	0	0.0320	0.0320	yes	yes

Table S3 — Complete Rashomon intervals (3/4)

Coordinatewise minimum and maximum gate weights among mixtures whose OOF MSE is within the stated relative tolerance of the optimum. All solver-success flags are checked by `verify_release.py`; KKT diagnostics are in `results/rashomon_kkt_audit.csv`.

Corpus	Expert	Tol.	Opt.	Min	Max	Width	Min OK	Max OK
SingMOS	BEATs	0.0050	0.1525	0.0661	0.2390	0.1729	yes	yes
SingMOS	CED	0.0050	0.2314	0.1235	0.3387	0.2152	yes	yes
SingMOS	ContentVec	0.0050	0.0761	0.0042	0.1480	0.1439	yes	yes
SingMOS	ERB	0.0050	4.68e-18	4.68e-18	0.0674	0.0674	yes	yes
SingMOS	RMVPE-c	0.0050	1.13e-17	1.13e-17	0.0371	0.0371	yes	yes
SingMOS	RMVPE-q	0.0050	4.92e-18	4.92e-18	0.0506	0.0506	yes	yes
SingMOS	Speaker	0.0050	0.0078	0	0.1098	0.1098	yes	yes
SingMOS	WavLM	0.0050	0.2156	0.1171	0.3141	0.1970	yes	yes
SingMOS	Whisper	0.0050	0.1036	0.0358	0.1713	0.1355	yes	yes
SingMOS	eGeMAPS	0.0050	0.2129	0.0960	0.3218	0.2258	yes	yes
TMHINT-QI	BEATs	0.0050	0.2710	0.0897	0.4515	0.3618	yes	yes
TMHINT-QI	CED	0.0050	0.2837	0.0970	0.4701	0.3731	yes	yes
TMHINT-QI	ContentVec	0.0050	0.0716	0	0.2320	0.2320	yes	yes
TMHINT-QI	ERB	0.0050	6.75e-19	6.75e-19	0.0459	0.0459	yes	yes
TMHINT-QI	RMVPE-c	0.0050	8.99e-18	8.99e-18	0.0494	0.0494	yes	yes
TMHINT-QI	RMVPE-q	0.0050	9.74e-18	9.74e-18	0.0436	0.0436	yes	yes
TMHINT-QI	Speaker	0.0050	0.0118	2.67e-16	0.1573	0.1573	yes	yes
TMHINT-QI	WavLM	0.0050	0.1697	1.12e-15	0.3603	0.3603	yes	yes
TMHINT-QI	Whisper	0.0050	0.1922	0.0407	0.3432	0.3025	yes	yes
TMHINT-QI	eGeMAPS	0.0050	0	0	0.1253	0.1253	yes	yes
BRSpeechMOS	BEATs	0.0100	0	0	0.0177	0.0177	yes	yes
BRSpeechMOS	CED	0.0100	1.67e-16	1.67e-16	0.0882	0.0882	yes	yes
BRSpeechMOS	ContentVec	0.0100	0.1532	0.0562	0.2503	0.1941	yes	yes
BRSpeechMOS	ERB	0.0100	0	0	0.0085	0.0085	yes	yes
BRSpeechMOS	RMVPE-c	0.0100	1.08e-16	1.08e-16	0.0585	0.0585	yes	yes
BRSpeechMOS	RMVPE-q	0.0100	2.44e-16	2.44e-16	0.0157	0.0157	yes	yes
BRSpeechMOS	Speaker	0.0100	0	0	0.1041	0.1041	yes	yes
BRSpeechMOS	WavLM	0.0100	0.3021	0.1958	0.4084	0.2126	yes	yes
BRSpeechMOS	Whisper	0.0100	0.0298	9.37e-16	0.1085	0.1085	yes	yes
BRSpeechMOS	eGeMAPS	0.0100	0.5149	0.3771	0.6266	0.2495	yes	yes

Table S3 — Complete Rashomon intervals (4/4)

Coordinatewise minimum and maximum gate weights among mixtures whose OOF MSE is within the stated relative tolerance of the optimum. All solver-success flags are checked by `verify_release.py`; KKT diagnostics are in `results/rashomon_kkt_audit.csv`.

Corpus	Expert	Tol.	Opt.	Min	Max	Width	Min OK	Max OK
BVCC	BEATs	0.0100	0.2003	0.0757	0.3249	0.2492	yes	yes
BVCC	CED	0.0100	0.2222	0.0833	0.3611	0.2777	yes	yes
BVCC	ContentVec	0.0100	0.0614	0	0.1834	0.1834	yes	yes
BVCC	ERB	0.0100	9.51e-18	9.51e-18	0.0254	0.0254	yes	yes
BVCC	RMVPE-c	0.0100	0	0	0.0293	0.0293	yes	yes
BVCC	RMVPE-q	0.0100	1.82e-18	1.82e-18	0.0268	0.0268	yes	yes
BVCC	Speaker	0.0100	0.0994	0	0.2269	0.2269	yes	yes
BVCC	WavLM	0.0100	0.1672	0.0319	0.3026	0.2707	yes	yes
BVCC	Whisper	0.0100	0.2495	0.1362	0.3627	0.2265	yes	yes
BVCC	eGeMAPS	0.0100	0	0	0.0560	0.0560	yes	yes
SingMOS	BEATs	0.0100	0.1525	0.0302	0.2747	0.2444	yes	yes
SingMOS	CED	0.0100	0.2314	0.0788	0.3828	0.3040	yes	yes
SingMOS	ContentVec	0.0100	0.0761	4.16e-17	0.1777	0.1777	yes	yes
SingMOS	ERB	0.0100	4.68e-18	4.68e-18	0.1108	0.1108	yes	yes
SingMOS	RMVPE-c	0.0100	1.13e-17	1.13e-17	0.0663	0.0663	yes	yes
SingMOS	RMVPE-q	0.0100	4.92e-18	4.92e-18	0.0777	0.0777	yes	yes
SingMOS	Speaker	0.0100	0.0078	0	0.1521	0.1521	yes	yes
SingMOS	WavLM	0.0100	0.2156	0.0763	0.3548	0.2785	yes	yes
SingMOS	Whisper	0.0100	0.1036	0.0078	0.1994	0.1916	yes	yes
SingMOS	eGeMAPS	0.0100	0.2129	0.0458	0.3655	0.3198	yes	yes
TMHINT-QI	BEATs	0.0100	0.2710	0.0146	0.5256	0.5110	yes	yes
TMHINT-QI	CED	0.0100	0.2837	0.0196	0.5468	0.5272	yes	yes
TMHINT-QI	ContentVec	0.0100	0.0716	0	0.2976	0.2976	yes	yes
TMHINT-QI	ERB	0.0100	6.75e-19	6.75e-19	0.0795	0.0795	yes	yes
TMHINT-QI	RMVPE-c	0.0100	8.99e-18	8.99e-18	0.0844	0.0844	yes	yes
TMHINT-QI	RMVPE-q	0.0100	9.74e-18	9.74e-18	0.0696	0.0696	yes	yes
TMHINT-QI	Speaker	0.0100	0.0118	2.67e-16	0.2176	0.2176	yes	yes
TMHINT-QI	WavLM	0.0100	0.1697	9.27e-15	0.4387	0.4387	yes	yes
TMHINT-QI	Whisper	0.0100	0.1922	0	0.4053	0.4053	yes	yes
TMHINT-QI	eGeMAPS	0.0100	0	0	0.1866	0.1866	yes	yes

Table S4 — Nested selection by corpus and budget (1/2)

Equal-system MSE in outer system-disjoint folds. Every rule uses the same cardinality k. Budgets {1,2,3,5} are the predeclared primary analysis; k={4,6,7,8,9,10} are secondary selection curves. Exhaustive is inner-validation search, never oracle selection on outer test systems. The full 160-row metric table also reports utterance MSE and system SRCC.

Corpus	k	Gate	Singleton	Shapley	Exhaustive
BRSpeechMOS	1	1.5432	1.5432	1.6283	1.5432
BRSpeechMOS	2	1.1520	1.1520	1.1520	1.1667
BRSpeechMOS	3	1.1427	1.1578	1.1543	1.1427
BRSpeechMOS	4	1.2561	1.1578	1.1506	1.2561
BRSpeechMOS	5	1.2174	1.1554	1.1112	1.2174
BRSpeechMOS	6	1.2174	1.2234	1.1829	1.2174
BRSpeechMOS	7	1.2174	1.1773	1.2336	1.2174
BRSpeechMOS	8	1.2174	1.2336	1.2336	1.2174
BRSpeechMOS	9	1.2174	1.2174	1.2174	1.2174
BRSpeechMOS	10	1.2174	1.2174	1.2174	1.2174
BVCC	1	0.3437	0.3338	0.3338	0.3338
BVCC	2	0.2820	0.2864	0.2918	0.2778
BVCC	3	0.2595	0.2654	0.2621	0.2585
BVCC	4	0.2497	0.2549	0.2514	0.2497
BVCC	5	0.2477	0.2474	0.2477	0.2477
BVCC	6	0.2469	0.2469	0.2469	0.2469
BVCC	7	0.2469	0.2469	0.2469	0.2469
BVCC	8	0.2469	0.2469	0.2469	0.2469
BVCC	9	0.2469	0.2469	0.2469	0.2469
BVCC	10	0.2469	0.2469	0.2469	0.2469

Table S4 — Nested selection by corpus and budget (2/2)

Equal-system MSE in outer system-disjoint folds. Every rule uses the same cardinality k . Budgets {1,2,3,5} are the predeclared primary analysis; $k=\{4,6,7,8,9,10\}$ are secondary selection curves. Exhaustive is inner-validation search, never oracle selection on outer test systems. The full 160-row metric table also reports utterance MSE and system SRCC.

Corpus	k	Gate	Singleton	Shapley	Exhaustive
SingMOS	1	0.4779	0.4779	0.4692	0.4779
SingMOS	2	0.4179	0.4105	0.4041	0.4041
SingMOS	3	0.3877	0.3842	0.3869	0.3921
SingMOS	4	0.3801	0.3844	0.3801	0.3793
SingMOS	5	0.3745	0.3819	0.3813	0.3739
SingMOS	6	0.3762	0.3814	0.3797	0.3749
SingMOS	7	0.3742	0.3797	0.3781	0.3742
SingMOS	8	0.3744	0.3796	0.3787	0.3744
SingMOS	9	0.3744	0.3770	0.3768	0.3744
SingMOS	10	0.3744	0.3744	0.3744	0.3744
TMHINT-QI	1	0.9986	0.9754	0.9754	0.9754
TMHINT-QI	2	0.9341	0.9272	0.9356	0.9363
TMHINT-QI	3	0.9158	0.9171	0.9138	0.9127
TMHINT-QI	4	0.9034	0.9042	0.9024	0.9034
TMHINT-QI	5	0.8993	0.9010	0.8993	0.9017
TMHINT-QI	6	0.8999	0.9011	0.8999	0.8999
TMHINT-QI	7	0.9001	0.9001	0.9001	0.9001
TMHINT-QI	8	0.9001	0.9001	0.9001	0.9001
TMHINT-QI	9	0.9001	0.9001	0.9001	0.9001
TMHINT-QI	10	0.9001	0.9001	0.9001	0.9001

Table S5a — Null: separate 1,000-draw redundancy simulation (1/1)

Complete numeric outcomes for this null family. Protocol notes remain verbatim in results/null_experiments.csv. These controls diagnose how much of the correlation–interaction pattern follows from regression geometry rather than speech-specific structure. This is a 1,000-draw geometric null simulation, distinct from the 100,000 node-label QAP permutations used for dyadic inference; it does not determine QAP p-values.

experiment	dataset	true rho corr vs neg	null mean	null sd	null p025	null p975	emp p null ge true	n null
permuted_redundan	BRSpeechMOS	0.3681	-0.3806	0.0413	-0.4621	-0.2957	0.0010	1.00e+03
permuted_redundan	BVCC	0.9431	-0.8460	0.0137	-0.8722	-0.8199	0.0010	1.00e+03
permuted_redundan	SingMOS	0.4432	-0.3411	0.0357	-0.4053	-0.2661	0.0010	1.00e+03
permuted_redundan	TMHINT-QI	0.9574	-0.8453	0.0098	-0.8664	-0.8288	0.0010	1.00e+03

Table S5b — Null: covariance matched gaussian (1/1)

Complete numeric outcomes for this null family. Protocol notes remain verbatim in results/null_experiments.csv. These controls diagnose how much of the correlation–interaction pattern follows from regression geometry rather than speech-specific structure.

experiment	dataset	n pairs	feasible	rho obs corr vs negl	rho matched null
covariance_matched_gaussian	BRSpeechMOS	45.0000	15.0000	0.3097	0.3016
covariance_matched_gaussian	BVCC	45.0000	41.0000	0.9578	0.9565
covariance_matched_gaussian	SingMOS	45.0000	42.0000	0.4279	0.8013
covariance_matched_gaussian	TMHINT-QI	45.0000	26.0000	0.9723	0.8852

Table S5c — Null: residual controls (1/1)

Complete numeric outcomes for this null family. Protocol notes remain verbatim in results/null_experiments.csv. These controls diagnose how much of the correlation–interaction pattern follows from regression geometry rather than speech-specific structure.

experiment	dataset	beta corr	beta mean	beta min
residual_controls	BRSpeechMOS	0.2342	0.9432	-0.1616
residual_controls	BVCC	0.3771	0.6984	-0.1368
residual_controls	SingMOS	0.0475	1.0525	-0.1467
residual_controls	TMHINT-QI	0.5501	0.5294	-0.1234

Table S5d — Null: full pipeline (1/1)

Complete numeric outcomes for this null family. Protocol notes remain verbatim in results/null_experiments.csv. These controls diagnose how much of the correlation–interaction pattern follows from regression geometry rather than speech-specific structure.

experiment	dataset	variant	observed nzero	observed count	null nzero mean	null nzero max	null count mean	null count max	null conv rate	n null
full_pipeline	BRSpeechMOS	eigclip	6.0000	5.0000	0	0	0	0	1.0000	5.0000
full_pipeline	BVCC	eigclip	4.0000	4.0000	0	0	0	0	1.0000	5.0000
full_pipeline	SingMOS	eigclip	3.0000	3.0000	0	0	0	0	1.0000	5.0000
full_pipeline	TMHINT-QI	eigclip	4.0000	4.0000	0	0	0	0	1.0000	5.0000
full_pipeline	BRSpeechMOS	higham	6.0000	5.0000	0	0	0	0	1.0000	5.0000
full_pipeline	BVCC	higham	4.0000	4.0000	0	0	0	0	1.0000	5.0000
full_pipeline	SingMOS	higham	3.0000	3.0000	0	0	0	0	1.0000	5.0000
full_pipeline	TMHINT-QI	higham	4.0000	4.0000	0	0	0	0	1.0000	5.0000

Table S6a — QAP dyadic inference (1/1)

This is the article's inferential QAP: 100,000 node-label permutations preserve both network structures. It is distinct from Table S5a's 1,000-draw geometric simulation and from the system bootstrap in Table S6b.

Corpus	Pairs	Observed rho	QAP p (2-sided)	QAP p (+)	Permutations
BRSpeechMOS	45	0.3681	0.0069	0.0069	100000
BVCC	45	0.9431	1.00e-05	1.00e-05	100000
SingMOS	45	0.4432	0.1243	0.0862	100000
TMHINT-QI	45	0.9574	1.00e-05	1.00e-05	100000

Table S6b — Interaction-correlation uncertainty (1/1)

Correlation of expert-prediction similarity with negative Shapley interaction. The test correlation interval resamples test systems while experts and fitted gates remain fixed; this is descriptive rather than full-refit uncertainty.

Corpus	Systems	OOF rho	OOF low	OOF high	Test rho	Test low	Test high	B
BRSpeechMOS	6	0.3243	0.0167	0.4801	0.5995	0.1161	0.7794	2000
BVCC	187	0.9040	0.8386	0.9310	0.9634	0.9107	0.9789	2000
SingMOS	35	0.3261	0.1233	0.7524	0.3354	0.1853	0.7444	2000
TMHINT-QI	97	0.9377	0.8087	0.9508	0.9568	0.8607	0.9743	2000

Artifact map and interpretation boundaries

Allocation: results/all_40_cells.csv contains point estimates, system intervals, and refit frequencies. Contextual utility: results/shapley_values.csv and results/interaction_values.csv contain exact coalition-game quantities. Near-optimal allocation: results/rashomon_intervals.csv contains every coordinate extreme; the KKT audit is in tables/supplementary_tables/.

Selection: results/nested_selection_metrics.csv contains every corpus $\times$ rule $\times$ budget outcome; the nested bootstrap contrasts are in tables/supplementary_tables/nested_selection_bootstrap.csv. Exhaustive means best inner-validation subset, not best outer-test subset.

Nulls: results/null_experiments.csv records four control families, and the interaction bootstrap is included in results/bootstrap_results.csv. QAP inference is in tables/supplementary_tables/qap_inference.csv. The matched Gaussian control reproduces much of the association, so no uniquely speech-specific law is claimed.

Scope: Shapley quantifies contextual utility under the declared player partition; it is not a universal selection rule. Singleton ranking was the attractive low-cost operating point in this benchmark, but no equivalence or universal encoder ranking is claimed. BRSpeechMOS system-level inference is exploratory because it has six systems.

Release identity: the version and repository URL are embedded above. The Git tag binds this compact snapshot to an immutable commit; no DOI field is displayed before Zenodo creates a real DOI.